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Usage of ResNet18 with CBAM Attention Mechanisms in Facial Emotion Recognition

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***Abstract*—Facial Emotion Recognition (FER) is a pivotal task in computer vision and is increasingly being powered by Artificial Intelligence (AI) techniques. These AI-driven applications span areas such as mental illness detection, biometrics, and psychological profiling. One of the most influential AI subsets employed in FER is deep learning, especially Convolutional Neural Networks (CNNs). The current state-of-the-art (SOTA) single-network classification accuracy on the FER-2013 dataset, achieved without additional training data, is attributed to the Visual Geometry Group (VGG) model, with a performance rate of 73.28%. In this work, we harness the capabilities of ResNet18 integrated with CBAM attention, achieving a test accuracy of 73.67% without the need for extra training data and also enhancing the convergence speed.**

Keywords*—*Facial Emotion Recognition, Artificial Intelligence, Convolutional Neural Networks, ResNet, CBAM, Computer Vision

I. Introduction

Facial Emotion Recognition (FER) has gained significant attention in computer vision and human-computer interaction fields. It aims to automatically identify emotional states from facial expressions, such as joy, anger, and sadness. The potential applications of FER span across various domains, from human-computer interaction and sentiment analysis to autonomous driving and medical diagnosis, offering opportunities to enhance user experience and address practical challenges [1].

Understanding emotions is essential as they play a pivotal role in human behavior and decision-making. Recognizing these emotional states is crucial for applications like mental health assessments, video conferencing, and humanoid-robotic interactions. While traditional methods, based on handcrafted features, have been prevalent, they often struggle with complex facial expressions [2]. The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), has revolutionized FER, allowing for better recognition of subtle patterns in facial images. Studies confirm the superior performance of CNNs in this domain [3]. For FER tasks, there is a famous public emotion recognition dataset, FER2013, which is provided by International Conference on Machine Learning (ICML) [4].

Nevertheless, traditional ResNet architectures have their limitations, especially in capturing global contextual information and handling different feature channels. The CBAM (Channel-wise Attention and Spatial Attention Module) attention mechanism addresses these challenges [5]. It refines the features by emphasizing relevant channels and regions within an image, enhancing the model's comprehension of object structures and semantics. CBAM has shown

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improvements in various computer vision tasks, from image classification to object detection.

In our research, we focus on enhancing the FER2013's prediction accuracy and convergence speed using ResNet with CBAM. We experiment with this architecture and achieve a test accuracy of 73.67%, marking, to the best of our knowledge, the highest accuracy for FER2013 using a singular network without additional data.

II. Related works

Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have been transformative in computer vision since AlexNet's victory in the ILSVRC 2012. Their influence in Facial Emotion Recognition (FER) has been substantial, transitioning the field from traditional machine learning methods to deep learning architectures [4].

A. CNNs based techniques

While early FER approaches relied on handcrafted features like Local Binary Patterns (LBP) and Gabor filters [6], the post-2012 era witnessed a preference for CNNs due to their inherent capability to learn spatially invariant features. Popular architectures such as VGG and ResNet have been tailored for emotion recognition with notable outcomes [7]. Moreover, specialized designs have been formulated to counteract challenges like varying illuminations and occlusions in FER [8].

Highlighting landmark CNN contributions: LeCun's 1989 LeNet primarily recognized handwritten digits [9]. AlexNet, introduced in 2012, featured eight layers, marking the deep learning boom [10]. VGG, designed by Oxford University's Visual Geometry Group, emphasized the importance of network depth, with VGG16 and VGG19 being its prominent versions. Utilizing VGG19, Khaireddin and Chen achieved a 73.28% accuracy [11].

B. Attention based techniques

Attention mechanisms, initially inspired by human cognition, have substantially improved deep learning models' performance. In the visual domain, Squeeze-and-Excitation Networks (SENet) introduced by Hu et al. recalibrate channel-wise features, elevating model efficacy [12]. Similarly, the Convolutional Block Attention Module (CBAM) by Woo et al. optimizes feature maps using channel and spatial attention [5].

U-Net, a seminal architecture by Ronneberger et al., laid the foundation for numerous image segmentation tasks [13]. The Transformer architecture, revolutionizing sequence modeling with its self-attention mechanism, was later adapted for visual tasks, as demonstrated by DETR for object detection [14].

Informed by these developments, we incorporate the CBAM mechanism into ResNet, inspired by Pramerdofer et al. and Woo et al. This fusion emphasizes salient regions in feature maps, enhancing emotion classification accuracy [3, 5].

III. Proposed method

A. ResNet18

In our proposed method, we employ the ResNet-18 as the basic architecture, an evolution of traditional deep convolutional networks designed to address the vanishing gradient problem in deep networks by introducing skip connections or shortcuts.

- **Initial Convolutional Layer:** The network begins with a convolutional layer that has 64 filters of size 3×3 and a stride of 1, ensuring the preservation of spatial dimensions (with a padding of 1). This layer is followed by batch normalization. The resulting feature map has dimensions of $64 \times 224 \times 224$.
- **Convolutional Blocks:**
  - **First Block:** Comprises 2 layers with 64 filters of size 3×3 in each layer. Both layers are equipped with batch normalization and ReLU activation.
  - **Second Block:** Consists of 2 layers, with 128 filters of size 3×3 in each layer. Like the first block, both layers have batch normalization and ReLU activation.
  - **Third Block:** Contains 2 layers with 256 filters of size 3×3 in each layer, both with batch normalization and ReLU activation.
  - **Fourth Block:** Includes 2 layers with 512 filters of size 3×3 in each layer, equipped with batch normalization and ReLU activation.

  In each block, the spatial dimensions are reduced by half if the input and output dimensions are unequal; otherwise, a 1×1 convolution is applied to match the dimensions.
- **Pooling Layer:** After the convolutional blocks, an average pooling operation of size 4×4 is performed, reducing the spatial dimensions to 1×1.
- **Fully Connected Layer:** The network concludes with a fully connected layer that has as many neurons as there are classes, in this case, 7. This layer produces the final classification scores for each class.

ResNet-18 is structured with a total of 18 convolutional layers segmented into blocks, one fully connected layer, and an average pooling layer. The incorporation of skip connections ensures smooth gradient flow, making the network suitable for deeper architectures. Batch normalization operations stabilize and expedite the learning process.

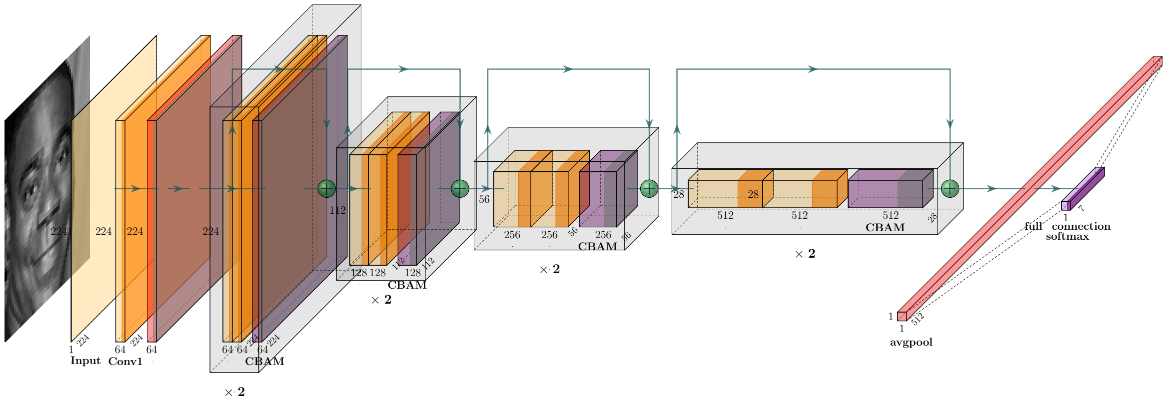

Figure 1. ResNet18 with CBAM architecture

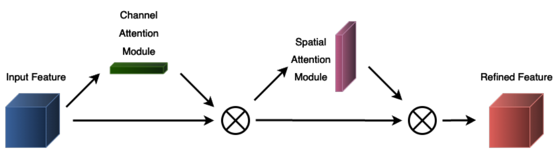

Figure 2. Overview of CBAM

B. *ResNet18 with CBAM*

In each residual block, we incorporated the CBAM module to enhance the model's representational capacity.

1) **CBAM Components**
   - **Spatial Attention**: Determines the importance of different spatial locations in a feature map by:
     - Computing both average and maximum values across channels.
     - Producing a spatial attention map using a 1×1 convolution.
     - Applying a sigmoid function to obtain values between 0 and 1.
   - **Channel Attention**: Assesses the significance of feature channels by:
     - Creating representations using Adaptive Average and Max Pooling.
     - Passing representations through a shared MLP with dimensionality reduction and restoration.
     - Applying a sigmoid function post-addition of outputs.
2) **CBAM in ResNet18's Basic Block**
   - Channel attention is applied post the third convolution (before adding the shortcut).
   - Spatial attention is applied next.
   - The attention-augmented result is then added to the shortcut connection.
3) **Benefits of CBAM Integration**
   - **Enhanced Representations**: Facilitates focus on critical features, potentially boosting performance.
   - **Adaptive Recalibration**: Allows the network to adjust feature maps emphasis based on input data.
   - **Efficiency**: CBAM adds minimal computational overhead.

   By incorporating the CBAM into the ResNet18 architecture, the model becomes more adaptive and can potentially produce more discriminative feature representations, which may aid in improving performance on tasks such as image classification.

IV. Experimental Results

In this section, we conduct experiments to test the overall performance of the model. We analyze the experimental setup, dataset characteristics, data augmentation techniques, hyperparameter selection, and model training conditions.

A. *Experimental setup*

In order to conduct our experiments, we utilized a computing system equipped with an NVIDIA GeForce RTX 3090 graphics card. This graphics card boasts approximately 24GB of VRAM, enabling us to effectively handle large-scale deep learning models and datasets. Throughout our experiments, we observed that our model required approximately 16GB of VRAM for training. These hardware constraints dictated the batch sizes and model complexities we could employ. We made necessary adjustments to ensure the smooth execution of our experiments under these hardware limitations.

B. FER-2013 dataset

Our experiment uses the Facial Expression Recognition 2013 (FER-2013) dataset. Comprising 35,887 grayscale 48x48 pixel images, it captures closed-mouthed frontal shots of faces across diverse ages, races, and backgrounds. The dataset categorizes faces into seven emotions: happy, sad, surprised, angry, fearful, disgusted, and neutral. Due to its relevance and broad use, FER-2013 serves as a benchmark in facial expression recognition research.

C. Data augmentation

In the data preprocessing phase, an extensive set of data augmentation techniques were employed to enhance the diversity of the training dataset and potentially improve the generalization capability of the model. The applied augmentation methods are delineated below:

1) **Mixup**: Linear interpolation between pairs of images and labels to enhance generalization and reduce overfitting.
2) **Color Jittering**: Adjusted brightness, contrast, and saturation to simulate lighting and color variations.
3) **Random Affine Transformation**: Applied translations to mimic image shifts.
4) **Horizontal Flip**: Introduced random horizontal flips for varied object orientations.
5) **Random Rotation**: Rotated images up to 10 degrees to account for varying angles.
6) **Perspective Transformation**: Introduced warping to simulate different viewpoints.
7) **Ten-crop Augmentation**: Cropped images into ten different regions for varied perspectives.
8) **Normalization**: Normalized pixel values using predefined statistics for consistent input scaling.
9) **Random Erasing**: Erased random patches to simulate occlusions.

These augmentations were systematically combined to expose the model to a vast array of image variations during training. Such a diverse training regimen is expected to fortify the model against overfitting and enhance its performance on unseen data.

D. Hyperparameters and training conditions

In our experiments, we employed a set of crucial hyperparameters and training conditions to ensure both the model's performance and the experiment's stability.

Our model was trained for a total of 300 epochs. Each training batch consists of 128 samples used for training and optimization. To effectively manage the learning rate, we adopted the CosineAnnealingLR learning rate scheduling strategy. Our initial learning rate (α_{start}) was set to 0.1, and we utilized the Stochastic Gradient Descent (SGD) optimizer with a momentum parameter of 0.9.

For enhancing the model's generalization performance, we enabled label smoothing with a smoothing parameter of 0.1. Furthermore, we applied Mixup data augmentation with a Mixup parameter of 1.0 to improve the model's robustness.

We performed multiple random crops (Ncrop) to augment the diversity of training samples.

The selection of these hyperparameters and training conditions was made to ensure the experiment's stability and the reliability of the results.

E. Performance Comparison

Table I
FER-2013 PUBLIC TESTSET BENCHMARK

Year	Model	Accuracy (%)
2016	CNN [15]	62.44
2018	GoogleNet [16]	65.20
2013	Human Accuracy [4]	65 ± 5
2019	VGG + SVM [17]	66.31
2017	Conv + Inception layer [18]	66.40
2013	Bag of Words [19]	67.40
2019	Attentional ConvNet [20]	70.02
2013	CNN + SVM [21]	71.20
2021	Resnet18 (ARM) [22]	71.38
2016	Inception [3]	71.60
2016	ResNet [3]	72.40
2021	VGG [11]	73.28
2023	**ResNet with CBAM**	**73.67**

Table I exhibits the classification accuracies of various models from prior related studies on the FER2013 public test set. Most of these methods surpass the estimated human performance of approximately 65.5%. The previously reported highest single-network accuracy, achieved without the use of additional training data, was 73.28%. In the present study, we attained the state-of-the-art accuracy of 73.67%.

Figure 3 displays the confusion matrix, where each row represents the predicted values of the target variable, and each column represents the actual values of the target variable. From the confusion matrix, it can be observed that our model performs best in the 'Happiness' class and 'Surprise' class, achieving accuracies of 0.89 and 0.83, respectively. In contrast, the model exhibits the lowest accuracy in the 'Fear' class, with only 0.57. This discrepancy may be attributed to the data imbalance within the FER-2013 dataset.

Figure 4 illustrates a comparison between the training and validation accuracies of ResNet and ResNet with CBAM. It is evident from the graph that ResNet with CBAM exhibits significantly faster convergence in both training and validation curves compared to the standard ResNet architecture. Moreover, it achieves

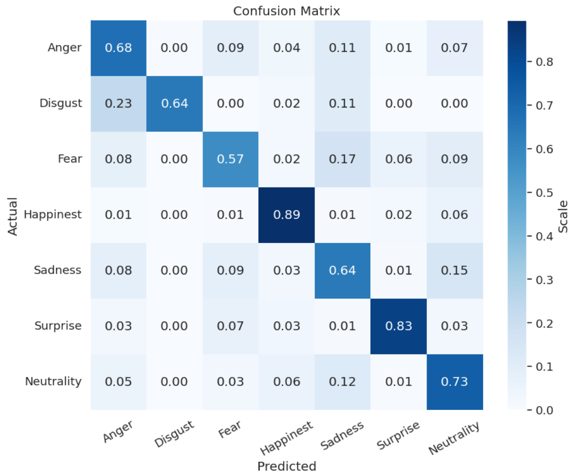

Figure 3. Confusion matrix of proposed ResNet18 with CBAM on the FER-2013 public test set

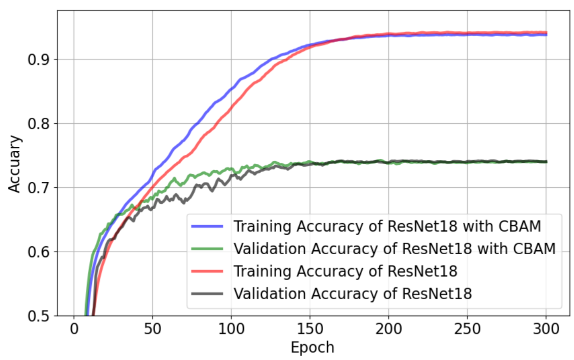

Figure 4. Training and Validation Accuracy Comparison of ResNet and ResNet with CBAM

a stable performance plateau in fewer epochs, suggesting a more efficient learning process. This improvement can be attributed to the integration of the CBAM mechanism, which enhances the model's ability to focus on informative features, adapt to different channels, and capture global contextual information. These findings underscore the efficacy of CBAM as a powerful enhancement to the ResNet framework, accelerating its convergence and improving its overall training dynamics.

V. Conclusion

In this work, we introduced the CBAM attention mechanism into the ResNet architecture, achieving state-of-the-art single-network classification accuracy on the FER2013 dataset without the use of additional training data. The CBAM attention mechanism is inserted at the end of each block in the BasicBlock of the model. Specifically, the CBAM module is applied after the features pass through the convolution and batch normalization layers, but before the activation function ReLU. The framework applies CBAM's attention mechanism to the feature representation of each basic block to improve the discriminative ability of each basic block and propagate these enhanced feature representations throughout the network. This helps to better capture key information in images and improve model performance. We conducted comparative experiments with other relevant studies on this dataset, demonstrating the superiority of our proposed model. Our model achieved a classification accuracy of 73.28%, which is the the state-of-the-art accuracy in our knowledge.

VI. Future Scope

Our results highlight the effectiveness of the CBAM attention mechanism in improving feature representation and the overall performance of deep neural networks. Furthermore, this work contributes to the ongoing efforts to advance the field of facial expression recognition. Future research directions may explore the integration of additional architectural enhancements and the application of our model in real-world scenarios.

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